Cadmium bioavailability from vegetable and animal-based foods assessed with in vitro digestion/caco-2 cell model.

BACKGROUND: Chronic dietary cadmium (Cd) exposure results in kidney dysfunction and decrease in bone mineral density. OBJECTIVE: To determine and compare the bioavailability of Cd from vegetable and animal-based foods. MATERIAL AND METHOD: Caco-2 cells were exposed to Cd in boiled pig kidney, ark shell, kale, raw kale, mixed boiled pig kidney with raw kale and CdCl2 after in vitro digestion. Then cellular Cd uptake from the digests and reference CdCl2 solution was measured by atomic absorption spectrometry. RESULTS: Cd bioavailability from animal-based foods was higher than that from vegetable-based foods. In addition, raw kale exhibited an inhibitory effect on Cd bioavailability when mixed with boiled pig kidney. However Cd in kale was increasingly absorbed after boiling. CONCLUSION: Cd binding to different molecular species, other food components in vegetable and animal-based foods, food combination, as well as cooking processes influenced the uptake of dietary Cd. A relative bioavailability factor accounted for the food matrix might be necessary for exposure assessment and consequently for estimation and prevention of the risk of dietary Cd.